

Project Design Phase-II

Technology Stack (Architecture & Stack)

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Project Name	Automated Network Request Management System
Maximum Marks	4 Marks

Technical Architecture Overview

The system is architected as a cloud-native enterprise application deployed on the ServiceNow Now Platform. It bridges the gap between end-user catalog submissions and direct backend hardware asset table persistence through a low-code automated workflow orchestration model.



Structural Breakdown:

- Frontend Tier (Local Client Interaction): Users interact with a clean Service Portal built using modern web standards. Client-side processing handles UI dynamic logic, instant context retrieval, and validation filters.
- Process Orchestration Engine (Cloud Logic): Triggers server-side logic pathways via ServiceNow Flow Designer and Workflow Studio. This engine manages operational gates, rules processing, and line-manager approval loops.
- Database Ledger Store (Persistence Layer): Securely seals validated data inputs directly into custom enterprise database schemas without manual text copy-pasting.

Table-1: Components & Technologies

S.No	Component	Descripon	Technology
1.	User Interface	Customer-facing Web portal interface where users search, select request opons, and view live stage progress badges.	ServiceNow Service Portal, HTML5, CSS3, JavaScript (ES6), Bootstrap Framework.
2.	Applicaon Logic-1	Client-side valida on scripts, session proper es query rules, and dynamic field visibility behaviors.	ServiceNow Client Scripts, Catalog UI Policies, GlideUser (g_user) API.
3.	Applicaon Logic-2	Server-side background execuon workflows that intercept user checkout paths and check rule condi ons.	ServiceNow Flow Designer, Workflow Studio Acons, Business Rules.
4.	Applicaon Logic-3	Cross-table rou ng calcula ons and rela onal approval mapping genera on handlers.	ServiceNow GlideRecord API, JavaScript Engine.
5.	Database	Local system data diconary structures holding system transac onal variables and workflow runme metrics.	ServiceNow Database Layer (built on MySQL/MariaDB founda on core).
6.	Cloud Database	Permanent, structural produc on rela onal table hosng the master infrastructure asset configura ons.	Custom System Schema Table (u_network_database).
7.	File Storage	Secure storage for transac onal audit history	ServiceNow Managed Cloud Aachment Storage (sys_aachment).

S.No	Component	Descripon	Technology
		logs and physical office floor plan document aachments.	
8.	External API-1	Iden ty contextual informaon query engine mapping incoming user sessions to corporate organiza onal directories.	ServiceNow internal GlideSystem session context lookup API.
9.	External API-2	Automated outbound no fica ons providing line managers with one-click decision-making funcons.	ServiceNow Cloud Mail Server Engine (SMTP/IMAP Noficaon Engine).
10.	Machine Learning Model	Oponal opmizaon rule set used to automacally assign fulfiller tasks based on agent workload queues.	ServiceNow Predicve Intelligence Engine (Oponal Similarity/Clustering Framework).
11.	Infrastructure (Server / Cloud)	High-availability cloud host layer ensuring mul-tenant isola on, data separa on, and secure runme boundaries.	ServiceNow Enterprise Cloud Plaorm-as-a-Service (PaaS) Instance.

Table-2: Applicaon Characteriscs

S.No	Characteris cs	Descripon	Technology
1.	Open-Source Frameworks	Integrat universally supported front-end grid layouts to ensure responsive UI display across various client browser viewports.	Bootstrap CSS Framework, AngularJS (ServiceNow Core UI framework stack wrapper).

S.No	Characteristics	Description	Technology
2.	Security Implementations	Data encryption definitions safeguarding active sessions, access control parameters, role profiles, and platform threat protection rules.	Advanced Encryption Standard (AES-256), Multi-Factor Authentication (MFA), Access Control Lists (ACLs), ServiceNow role-based access management.
3.	Scalable Architecture	Logical layer breakdown strategy that isolates user presentation elements from backend ledger transactions to manage fluctuating query traffic.	Multi-tier Cloud Platform-as-a-Service (PaaS) framework, microservices-inspired multi-node cloud application nodes.
4.	Availability	Resilient architectural planning guaranteeing zero service interruptions via continuous redundancy paths and fallback cluster instances.	ServiceNow Advanced High Availability (AHA) Architecture, geo-redundant automated data center replication.
5.	Performance	Computational tuning to enforce minimal form load delays, sub-second execution speeds, and background workflow balancing processing.	ServiceNow Client-Side Cache Optimization, Indexed Table fields on u_network_database, Asynchronous Background Workers.